

MICROSOFT POWERED HACKATHON

Microsoft Innovate 2026

Innovate · Build · Empower**Team Name***The Lookouts***Team ID***410***Problem Statement Title***10,000 Reviews, No Time to Read Them***Theme / Bucket***Smart Assistants and Chatbots***Team Leader***Vishesh Aggarwal, S24CSEU2521, aggarwalvishesh0@gmail.com***ROUND 1****Idea Submission****InSight**

REAL FEEDBACK. REAL INSIGHTS.

Submission deadline: 27th Sept 2026

Team Details

A passionate team building AI for real-world impact.



Round 1 – Idea Submission

#	S. No.	Full Name	Enrollment No.	Programme & Year	Email	Role in Team
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Problem Statement



Round 1 – Idea Submission

Valuable customer insights are buried in massive volumes of unstructured feedback.

01



What is the problem?

Product teams struggle to turn high-volume customer feedback into clear, trustworthy insights.

- **10,000+** reviews, survey responses, and support comments can accumulate quickly.
- Manual reading and categorization are **slow, inconsistent, and difficult to scale**.
- Recurring complaints, sentiment shifts, and emerging themes are **easy to overlook**.



02



Who faces it?

Teams that rely on customer feedback to understand experience and guide decisions.

- **Product Managers** — identify friction and prioritize improvements.
- **Customer Support** — understand recurring complaints across conversations.
- **Business Analysts** — quantify qualitative feedback and communicate trends.



Product Managers



Customer Support



Business Analysts

03



Why do current options fall short?

Feedback is abundant, but extracting reliable, explainable patterns remains challenging.

- **Traditional BI tools** excel at structured metrics but need preparation to analyze unstructured text.
- **Keyword-based methods** miss semantic similarities (e.g., “app crashed” and “software keeps closing”).
- **Basic sentiment summaries** often lack theme-level trends and links back to real review examples.

Keyword Matching

Misses variations and context

crash

error

bug



Meaning-Based Themes

Groups similar feedback intelligently

app crashed

keeps closing

won't open

suddenly shuts down

Proposed Solution



Round 1 – Idea Submission



Our idea in one line

InSight transforms thousands of customer reviews into **clear themes, sentiment trends, and evidence-backed insights**—visualized in its own dashboard and Power BI.



How it solves the problem

1

Semantic theme discovery

Groups feedback by meaning, not just keywords.

2

Sentiment & complaint analytics

Reveals recurring complaints, sentiment patterns, and emerging topics.

3

Traceable insights

Connects every theme and chart to real customer review examples.



What makes it different

1

ML-powered understanding

Dense embeddings and NLP uncover semantic patterns across large review sets.

2

Two ways to explore

Interactive InSight dashboard plus Power BI reporting for shareable, customizable visualizations.

3

Trustworthy analytics

Validated sentiment, privacy-aware processing, and clear links to source feedback.



Reviews

User feedback



ML Analysis

Themes • Sentiment



Insights

Trends • Examples

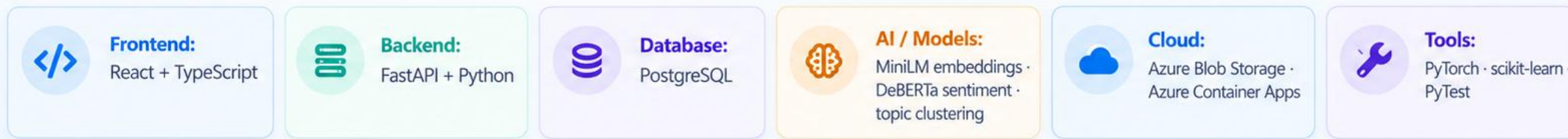


InSight + Power BI

Dashboard • Reports

Technical Approach

Planned tech stack



How the system will work



What we have started working on

- FastAPI v2 migrated with tested overview and themes endpoints
- Precomputed embeddings and SHA-256 cache for repeatable, faster analysis
- Fine-tuning and confidence-calibration notebooks prepared; sample datasets ready



Impact & Next Steps



Round 1 – Idea Submission



Expected Impact

- Turn high-volume reviews into **traceable issue themes**.
- Pilot target: **reduce manual triage time by 80%**.
- Surface emerging **high-severity issues earlier** with evidence-linked insights.



Feasibility

- **FastAPI v2** and core overview/theme endpoints are implemented and tested.
- **Cached embeddings** and **sample datasets** support repeatable analysis.
- Team skills span ML, Python APIs, frontend, and data engineering.



Plan for the Next Round

- Validate **clustering**, sentiment, and severity against labeled examples.
- Add **confidence gates**, **duplicate checks**, and **human-approved Jira drafts**.
- **Demo**: feedback → emerging issue → evidence-backed ticket; measure triage time.



References & Data Sources

- *Anonymized sample feedback*: SaaS and D2C product reviews.
- Sentence-BERT (Reimers & Gurevych, 2019); DeBERTa (He et al., 2021).
- **BERTopic** (Grootendorst, 2022); Jira issue workflow documentation.

